

CASE-18

Network Forensics Investigation Report

SSLoad + Cobalt Strike C2 Beacon Analysis

Field	Value
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Date	August 16, 2026
PCAP Source	Malware Traffic Analysis -- 2024-04-18
Environment	REMnux Ubuntu 24.04 Zeek 8.2.1 RITA v5.1.2 JupyterLab 4.6.3
Classification	TLP:WHITE -- For Educational Use
Repository	github.com/ronankongala/zeek-network-forensics-lab

Executive Summary

This report documents the analysis of a real-world malware PCAP capturing an SSLoad infection with follow-on Cobalt Strike DLL activity from April 18, 2024. Network traffic from victim host 10.4.18.169 was analyzed using Zeek 8.2.1 for structured log generation and RITA v5.1.2 for probabilistic beacon scoring. Analysis confirmed active C2 communication to 85.239.53.219 on port 80 over HTTP, with 11 recorded connections spanning a total of 5,087 seconds. RITA assigned a beacon score of 0.504 and auto-tagged the traffic with rare_signature 'SSLoad/1.1'. Mean inter-connection interval was 477 seconds with low byte counts per callback, consistent with a configured C2 sleep timer. Secondary suspicious channels included api.openweathermap.org and t.me (Telegram), both scoring 0.682 on RITA's beacon scale. Three Jupyter notebooks document the full detection methodology across conn.log analysis, DNS query profiling, and beacon interval visualization.

1. Technical Findings

1.1 Zeek Log Summary

Zeek 8.2.1 ran in standalone mode against the PCAP, generating 17 structured log files totaling 220KB. kerberos.log and ldap.log alongside C2 traffic indicate post-exploitation Active Directory enumeration.

Log File	Size	Significance
conn.log	27K	All connections -- primary beacon detection source
http.log	81K	273 HTTP requests including SSLoad C2 callbacks
dns.log	19K	73 DNS queries including C2 dead drop domains
ssl.log	12K	TLS connections -- Cobalt Strike HTTPS beacon candidates
files.log	8.3K	DLL payload transfers detected
kerberos.log	2.1K	AD authentication -- post-exploitation lateral movement indicator
ldap.log	2.8K	AD enumeration activity following initial compromise
dce_rpc.log	8.3K	Remote procedure calls -- lateral movement indicator

1.2 C2 Beacon Identification -- conn.log

Sorting conn.log by connection duration surfaced 85.239.53.219:80 as the primary C2 server. 11 connections were recorded over approximately 80 minutes. Three connections exceeded 17 minutes in duration with low byte counts -- the hallmark of a C2 beacon with a sleep timer.

Timestamp (Unix)	Duration (s)	Duration (min)	Bytes	State
1713465805.964	1982.689	33.0	29,967	SF
1713467808.656	1992.108	33.2	30,071	SF
1713469820.768	1034.494	17.2	15,600	SF
1713466385.888	6.743	0.1	118	SF
1713466987.722	6.686	0.1	118	SF
1713467580.694	21.688	0.4	118	SF
1713468183.700	6.667	0.1	118	SF
1713468783.833	6.754	0.1	118	SF
1713469390.769	6.672	0.1	118	SF
1713469985.748	13.706	0.2	118	SF
1713470580.738	9.769	0.2	118	SF

The alternating pattern of long sustained connections and brief 118-byte check-ins indicates the beacon cycling between active tasking sessions and idle polling. Mean inter-connection interval across all 11 connections was 477 seconds (~8 minutes).

1.3 RITA Beacon Scoring Results

RITA v5.1.2 scored all external connections for beaconing regularity. Scores above 0.5 warrant investigation. 85.239.53.219 received the rare_signature:SSLoad/1.1 modifier, confirming RITA's threat intel feeds identified the SSLoad user agent in HTTP traffic.

Destination	Score	Conns	Duration	Severity	Modifier
t.me	0.682	9	190s	Low	--
api.openweathermap.org	0.682	9	2.16s	Low	--
login.microsoftonline.com	0.590	7	6.33s	None	--
85.239.53.219	0.504	11	5,087s	Low	rare_sig: SSLoad/1.1
settings-win.data.microsoft.com	0.196	4	2.16s	None	--

1.4 DNS Analysis

dns.log contained 73 queries. wpad.partridgecliff.com was the top queried domain -- WPAD queries indicate the malware probing for proxy configurations to route C2 through legitimate channels. api.openweathermap.org and t.me align with RITA's beacon findings.

Domain	Count	Assessment
wpad.partridgecliff.com	Highest	WPAD probe -- proxy discovery for C2 evasion
api.openweathermap.org	High (9)	Suspected dead drop resolver / fallback C2
t.me	High (9)	Telegram -- suspected exfil or fallback C2 channel
login.microsoftonline.com	Med (7)	Likely legitimate Microsoft authentication
v10.events.data.microsoft.com	Medium	Microsoft telemetry -- likely benign
partridge-dc.partridgecliff.com	Medium	Domain controller lookup -- AD enumeration

2. MITRE ATT&CK; Mapping

Technique	Name	Evidence
T1071	Application Layer Protocol	SSLoad C2 over HTTP port 80 to 85.239.53.219; 273 HTTP requests in http.log
T1071.004	DNS C2	High-frequency DNS queries to api.openweathermap.org and t.me; RITA beacon score 0.682 on both
T1008	Fallback Channels	t.me (Telegram) and api.openweathermap.org queried in parallel with primary C2 -- configured fallback channels
T1095	Non-Application Layer Protocol	ssl.log records TLS sessions to multiple external IPs alongside plaintext HTTP C2 -- Cobalt Strike HTTPS beacon
T1557	Adversary-in-the-Middle	wpad.partridgecliff.com is the top DNS query -- WPAD poisoning attempt to intercept and redirect proxy traffic
T1018	Remote System Discovery	partridge-dc.partridgecliff.com DNS lookup -- domain controller enumeration post-compromise

3. Indicators of Compromise

IOC Type	Value	Context
IP Address	85.239.53.219	SSLoad C2 -- port 80 HTTP -- 11 connections -- beacon score 0.504
IP Address	10.4.18.169	Victim host -- source of all C2 callbacks
Domain	api.openweathermap.org	Dead drop resolver -- beacon score 0.682 -- 9 connections
Domain	t.me	Telegram endpoint -- fallback C2 -- beacon score 0.682
Domain	wpad.partridgecliff.com	WPAD probe -- proxy discovery -- highest DNS query count
Domain	partridge-dc.partridgecliff.com	Domain controller -- AD enumeration target
User Agent	SSLoad/1.1	SSLoad malware HTTP user agent -- flagged by RITA rare_signature
Port	80/tcp	Primary C2 port -- plaintext HTTP
Port	445/tcp	SMB -- lateral movement to 10.4.18.4 (212s, 14KB)
Beacon Interval	477 seconds	Mean inter-connection interval -- Cobalt Strike sleep timer
Beacon Jitter	~25%	Interval std deviation indicates configured jitter value
File Type	DLL	Cobalt Strike beacon delivered as DLL (files.log 8.3KB)

4. Detection Recommendations

4.1 Zeek + RITA Deployment

Deploy Zeek as a network tap on the perimeter or at east-west chokepoints to generate structured TSV logs. Feed logs to RITA on a rolling 24-hour import schedule: 'rita import --database daily --rolling --logs /path/to/zeek/logs'. Alert on any beacon score above 0.5 for analyst review. The rare_signature modifier provides automatic malware family tagging when known user agents appear in HTTP traffic.

4.2 Sigma Rules

```
title: SSLoad C2 Beacon via HTTP
id: case18-001
status: experimental
description: Detects SSLoad C2 via long-duration low-byte HTTP connections
logsource:
  product: zeek
  service: conn
detection:
  selection:
    proto: tcp
    id.resp_p: 80
    duration|gt: 600
    orig_bytes|lt: 5000
  condition: selection
level: high
tags:
  - attack.t1071
```

```
title: High-Frequency DNS to Non-Corporate Domain
id: case18-002
status: experimental
description: Detects repeated DNS queries suggesting DGA or C2 dead drop
logsource:
  product: zeek
  service: dns
detection:
  selection:
    qtype_name: A
  timeframe: 10m
  condition: selection | count(query) by query > 8
level: medium
tags:
  - attack.t1071.004
```

4.3 Network Controls

Control	Implementation
Block WPAD externally	DNS firewall rule blocking wpad.* lookups leaving the network perimeter
Restrict Telegram egress	Block t.me and *.t.me at the proxy layer -- not a legitimate business application
Monitor port 80 outbound	Alert on connections longer than 10 minutes over plain HTTP to non-whitelisted IPs
RITA rolling import	Automate daily Zeek log ingestion; page on any beacon score above 0.5
User agent inspection	Deploy HTTP inspection at the proxy to alert on unknown or rare user agent strings
SMB monitoring	Alert on SMB connections to unexpected internal hosts -- lateral movement indicator

5. Analysis Methodology

Phase	Tool	Output
1 -- Environment Setup	Zeek 8.2.1, RITA v5.1.2, JupyterLab 4.6.3	Lab on REMnux Ubuntu 24.04
2 -- PCAP Acquisition	wget, unzip	SSLoad + Cobalt Strike PCAP (6.4MB, MTA 2024-04-18)
3 -- Log Generation	zeek -r pcapfile	17 structured log files (220KB total)
4 -- Beacon Scoring	rita import, rita view	Beacon scores for all external connections
5 -- Threat Hunting	Jupyter + pandas + matplotlib	3 notebooks: conn, DNS, beacon interval analysis
6 -- Reporting	Python ReportLab	This PDF -- ATT&CK; mapped, IOC table, detection rules

6. References

- Malware Traffic Analysis 2024-04-18: malware-traffic-analysis.net/2024/04/18/
- MITRE ATT&CK; T1071: attack.mitre.org/techniques/T1071/
- MITRE ATT&CK; T1008: attack.mitre.org/techniques/T1008/
- MITRE ATT&CK; T1071.004: attack.mitre.org/techniques/T1071/004/
- MITRE ATT&CK; T1095: attack.mitre.org/techniques/T1095/
- RITA v5 Repository: github.com/activecm/rita
- Zeek Network Security Monitor: zeek.org
- Lab Repository: github.com/ronankongala/zeek-network-forensics-lab